ModelIt!

Teacher's Guide

Atoms Rearrange: Modeling Chemical Reactions

Conservation of Mass in Action

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-PS1-2, MS-PS1-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS1-2, MS-PS1-5

Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how reactant concentration and temperature affect reaction rate and product formation while total mass remains conserved.
Disciplinary Core Idea	**PS1.B Chemical Reactions** — Substances react chemically in characteristic ways. In a chemical process, the atoms that make up the original substances are regrouped into different molecules, and these new substances have different properties from those of the reactants.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students demonstrate that matter is conserved during chemical reactions. The total number of atoms of each element is the same before and after the reaction, even though atoms rearrange.

CAST Testing Alignment

MULTIPLE CHOICE:

A student mixes 50 grams of baking soda with 100 grams of vinegar in an open beaker. After the reaction stops, the beaker and contents weigh 140 grams. What best explains the missing 10 grams?

CONSTRUCTED RESPONSE:

A student claims that burning a log destroys matter because the ashes weigh less than the original log. Using the law of conservation of mass and your model of chemical reactions, explain why this claim is incorrect. Include what happens to the atoms in the log.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Conservation of mass is one of the most fundamental laws in all of science. Antoine Lavoisier established it in the 1770s, earning him the title 'Father of Modern Chemistry.' Every chemical reaction in the universe obeys this law without exception.

Lavoisier's Discovery

Before Lavoisier, scientists believed that burning destroyed matter. Lavoisier designed careful experiments with sealed containers and precise measurements, proving that the total mass before and after a reaction is identical. He showed that burning requires oxygen from the air and produces gases. No mass is lost; it just changes form.

Atoms Rearrange, Not Transform

In a chemical reaction, atoms do not change identity. A carbon atom stays a carbon atom. An oxygen atom stays an oxygen atom. What changes is how atoms are connected to each other. In burning wood, carbon atoms bonded to hydrogen and oxygen in cellulose molecules break apart and rebond with oxygen from the air to form CO2 and H2O. Same atoms, new arrangement.

Why Reactions Seem to Destroy Matter

Reactions appear to destroy matter because products are often gases that disperse into the air. Burning wood produces CO2 and water vapor that float away, leaving only ash (minerals that did not combust). In an open system, the gases escape and the remaining solid weighs less. In a sealed system, all products stay, and the mass is identical.

Reaction Rate Factors

Temperature increases reaction rate because molecules move faster and collide more often with more energy. Higher concentration increases rate because more molecules are packed together, increasing collision frequency. Surface area matters because reactions happen at surfaces where molecules can contact each other. Catalysts lower the energy barrier for reactions without being consumed.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify reactant concentration and temperature as external inputs, and reaction rate, product formation, and total mass as internal properties of the reaction system.

E **Establish Relationships**

Students determine that both reactant concentration and temperature positively drive reaction rate, reaction rate positively drives product formation, and reaction rate negatively affects reactant concentration (reactants get used up). Total mass remains constant.

V **Visualize & Model**

Students build a model showing reactant and product lines as inverse curves with a flat total mass line, demonstrating conservation of mass.

E **Evaluate Outcomes**

Students run fast vs. slow reaction scenarios and observe that while the RATE changes, the total mass is always conserved, and the same final amounts of products form.

R **Revise & Extend**

Students add surface area or catalyst presence to explore additional factors that affect reaction rate without changing total mass.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"When you burn a match, the wood seems to vanish. But here is a mind-bending fact: NOT ONE ATOM was destroyed. So where did it all go?"*

Watch For

Light a match (or show video). Ask students: Where did the wood go? Capture responses.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to prove that matter is NEVER created or destroyed. Atoms just rearrange."*

Watch For

Have students read objectives. Pre-teach 'reactant,' 'product,' and 'conservation of mass.'

Slide 3: Big Question

3 min

Say: *"A 10-pound log burns down to a few ounces of ash. Where did the other 9+ pounds go?"*

Watch For

Think-pair-share: Does burning destroy matter? Students write predictions before discussion.

Slide 4: LEVER Framework

2 min

Say: *"Our model will show two lines that are perfect mirror images, and one line that NEVER moves. That flat line is the key."*

Watch For

Preview the 5 components. Tease that one variable (total mass) will stay perfectly constant.

Slide 5: Activity 1: Components

8 min

Say: *"What can we control in a chemical reaction? What responds? What stays the same no matter what?"*

Watch For

Guide sorting. Emphasize that Total Mass is internal but CONSTANT. This is the conservation law.

Slide 6: Activity 2: Connections

8 min

Say: *"As reactants go DOWN, products go UP by the exact same amount. They are perfect inverses. This is conservation."*

Watch For

Use physical model: start with 20 red blocks (reactants), convert to blue blocks (products) one at a time. Count total at each step.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run a fast reaction and a slow reaction. The rates change, but watch that total mass line."*

Watch For

Students predict graphs first. Run fast and slow scenarios. Overlay graphs. Point to the flat mass line.

Slide 8: Discoveries

5 min

Say: *"Lavoisier proved this 250 years ago and it has NEVER been violated. Every reaction in the universe conserves mass."*

Watch For

Connect to the burning log question. Explain that gases (CO_2, H_2O vapor) carry the missing mass away.

Slide 9: STEM Challenge

12 min

Say: *"Prove it yourself. Design an experiment that captures ALL products and shows mass is conserved."*

Watch For

Distribute materials (bags, bottles, scales). Groups design, execute, and measure sealed reactions.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Reactant Concentration, Temperature
- **INTERNAL:** Reaction Rate, Product Formation, Total Mass

Relationships

- **Reactant Concentration to Reaction Rate:** POSITIVE (+) — More reactant molecules means more collisions per second, driving a faster reaction rate. As reactants are consumed, the rate naturally slows.
- **Temperature to Reaction Rate:** POSITIVE (+) — Higher temperature makes molecules move faster, collide more often, and collide with more energy. This speeds up the reaction rate significantly.
- **Reaction Rate to Product Formation:** POSITIVE (+) — A faster reaction rate means products are being created more quickly. The product line rises faster when reaction rate is high.
- **Reaction Rate to Reactant Concentration:** NEGATIVE (-) — As the reaction proceeds, it consumes reactants. Higher reaction rate means reactants are consumed faster, causing concentration to drop. This creates a self-slowing effect.
- **Total Mass: CONSTANT:** CONSERVATION — Total Mass never changes because atoms are neither created nor destroyed. As reactant mass decreases, product mass increases by exactly the same amount. The sum is always constant.

Key Discoveries

- Reactant and product lines are INVERSE: as one goes down, the other goes up by exactly the same amount
- Total Mass stays perfectly flat throughout the reaction, proving conservation of mass
- Higher temperature increases reaction rate because molecules collide more frequently and with more energy
- As reactants are consumed, the reaction rate SLOWS because there are fewer molecules available to collide
- Atoms are not created or destroyed. They just rearrange from one molecular configuration to another

Multiple Choice Post-Assessment Answer Key

1. In the chemical reaction model, what happens to the Total Mass line when reactants convert to products?

Correct: C) Total Mass stays perfectly flat because atoms are neither created nor destroyed, they just rearrange

Correct! Conservation of mass means the total mass never changes during a chemical reaction. As reactant mass decreases, product mass increases by exactly the same amount. The total is always constant because atoms are rearranging, not appearing or disappearing.

2. A student burns a 200-gram log in an open fire and the remaining ash weighs only 15 grams. Does this violate conservation of mass?

Correct: B) No, the missing 185 grams escaped as carbon dioxide gas and water vapor into the atmosphere

Correct! Burning is a chemical reaction between the log's carbon and hydrogen atoms with oxygen from the air. The products are CO2 gas and water vapor, which float away into the atmosphere. The ash is only the minerals that did not combust. If you could capture and weigh all the gases, the total mass would equal the original log plus the oxygen consumed.

3. Why does the reaction rate slow down as a chemical reaction proceeds, even if the temperature stays constant?

Correct: C) As reactant molecules are consumed, fewer molecules remain to collide, so collisions become less frequent

Correct! Reaction rate depends on how often reactant molecules collide. As reactants are consumed and converted to products, fewer reactant molecules remain in the system. Fewer molecules means fewer collisions per second, which slows the reaction. This is a self-slowing system, similar to how cooling curves flatten.

4. How does increasing temperature affect a chemical reaction?

Correct: B) Higher temperature makes molecules move faster, collide more often, and collide with more energy, increasing the reaction rate

Correct! Temperature is a measure of molecular motion. Higher temperature means molecules move faster, which increases both the frequency and the energy of collisions. More frequent, more energetic collisions mean more successful reactions per second, increasing the reaction rate.

5. In the model, reactant concentration and product formation are described as 'mirror images' of each other. What does this mean and why does it happen?

Correct: B) As reactant concentration decreases by a certain amount, product formation increases by exactly the same amount, because atoms are rearranging rather than being created or destroyed

Correct! Since atoms are neither created nor destroyed, every gram of reactant that disappears reappears as a gram of product. The reactant line goes down while the product line goes up by the same amount. They are perfect inverses because conservation of mass guarantees that what is lost from one side is gained by the other.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Burning destroys matter"	Burning is a chemical reaction between fuel and oxygen that produces new substances (CO_2, H_2O) as gases. These gases float away, making it LOOK like matter disappeared. In a sealed container, the mass before and after burning is identical. Not one atom is destroyed.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavier products mean mass was created"	When iron rusts, it gains mass because it combines with oxygen from the air. The oxygen atoms were already there in the atmosphere. No mass was created; it was transferred from the air to the iron. The total mass of the system (iron + air) stays the same.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Chemical reactions create new atoms"	Atoms are NEVER created or destroyed in chemical reactions. Only nuclear reactions (fusion, fission) change atom identities. In chemistry, atoms simply rearrange their connections. Carbon stays carbon. Oxygen stays oxygen. The only thing that changes is which atoms are bonded to which.